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SCUOLA DI INGEGNERIA INDUSTRIALE
E DELL'INFORMAZIONE

Laurea Magistrale in Biomedical Engineering - Ingegneria Biomedica

Needle Optimization and Trajectory Planning for Autonomous Anastomosis Suturing

Medical Robotics and Technologies for Computed Aided Surgery Laboratory

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Academic year: 2024-2025

1. Introduction

Anastomosis is the process of joining branching networks of tubes into a continuous structure. In surgery, this commonly occurs in the circulatory and digestive systems and, less frequently, in the genitourinary tract. The most common types are vascular and intestinal anastomoses. Vascular anastomosis joins two blood vessels (arteries or veins), while an intestinal or bowel anastomosis reconnects parts of the intestines. Surgeons can reconnect these body channels in various ways:

- **End-to-end anastomosis**: Two open ends are connected together to form a seamless tube.
- **Side-to-side anastomosis**: Two channels are connected side-by-side, and their ends are closed.
- **End-to-side anastomosis**: One open end is connected to the side of another channel, and the open end of that channel is closed.

End-to-end anastomosis (ETEA) is a frequent microsurgical technique, particularly in hand surgery for repairing collateral arteries and nerves (wound management, multi-tissue lesions, digital replantation, etc.). Poor execution of this technique can lead to failure of the patency test, in situ thrombosis, or aneurysm formation. It is a time-consuming and technically demanding procedure, making the optimization of its steps highly valuable [1].

One widely used approach is the Harashina end-to-end suture technique, shown in Figure 1, which is based on biangulation to prevent the sutures from being transfixed. This technique is typically performed on vessels with diameters between 0.7 and 2 mm. Eight equidistant sutures are placed along the circular wall while ensuring that the lumen remains open to prevent transfixion [2]. However, all manual suturing techniques share a common limitation: they require sufficient vessel length and significant operative time.

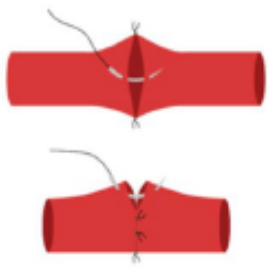


Figure 1: Harashina anastomosis technique

A surgeon's skill directly impacts patient outcomes, making surgical training essential for the progressive mastery of various technical abilities, including precise tissue handling, dissection, and suturing. Effective suturing techniques are crucial across all surgical specialties and platforms, including robotic-assisted surgery (RAS). This approach has become a key advancement in minimally invasive surgery (MIS) due to its significant potential. In RAS, the surgeon operates from a console using master controls to direct the surgical instruments. This technique offers several advantages: it enhances visualization of the surgical field, leading to greater precision; it increases the range of motion and the surgeon's dexterity, thereby reducing potential complications during surgery. Robotic-assisted suturing has well-documented clinical benefits, including faster patient recovery. However, achieving optimal results still depends on the surgeon's ability to operate the technology effectively [3].

To objectively assess and refine robotic-assisted suturing techniques, depending on surgeon's skills, Taseen et al. [4] developed the *End-to-End Assessment of Suturing Expertise* (EASE), which defines the key sub-skills involved in robotic suturing and establishes optimal suturing parameters. The procedure is systematically divided into 18 sub-skills, each with well-defined geometric and procedural characteristics.

Key findings from EASE indicate that:

- The needle should enter the tissue plane perpendicularly or at an angle between 45° and 90°.
- The needle-driving motion must be continuous and smooth, following the natural curvature of the needle to minimize tissue trauma.
- The depth of penetration should be sufficient to involve all vessel layers appropriately.
- Stitches should be evenly spaced and proportionally distributed without gaps, ensuring alignment of tissue ends and suture bites.

These optimized suturing principles, validated for robotic-assisted surgery, are not only applicable to human-guided procedures but also serve as the foundation for developing fully autonomous robotic suturing systems. As the field moves toward greater automation in surgical procedures, these established criteria will play a crucial role in training robotic systems capable of performing precise and reliable autonomous suturing.

Currently, there is growing interest in the study of autonomous suturing [5]. Robotic surgical assistants have demonstrated promising capability to unburden medical staff by automation of camera steering. Similarly, robots could be programmed to perform all or part of the surgery with or without presence of an expert. Automation, furthermore, reduces the cognitive loads on surgeons. This is beneficial, in particular, for time-consuming and repetitive surgical tasks such as suturing, granting reduced procedural time and risk, and increasing clinical output and reliability [6]. Although research in medical robotics is advancing, fully autonomous anastomosis remains an open challenge, with only a limited number of studies available in the literature. Recent work has explored the possibility of a semi-autonomous approach to this procedure [7], while the goal of this study is to develop a trajectory generation algorithm to enable fully autonomous suturing.

Suturing is a compound task consisting of subtasks such as positioning the needle, biting the tissue, driving the needle through the tissue and then tying a knot in order to close the wound. This study focuses on the development of a path planning algorithm that can guide the needle while minimizing tissue trauma. To achieve this, two main autonomous suturing strategies have been proposed in the literature: *Fixed-Centre Motion (FCM)* and *Moving-Centre Motion (MCM)*.

- In the FCM approach, the needle follows a circular trajectory around a fixed center while performing the suture.
- In contrast, MCM allows the needle to translate or rotate freely, meaning its geometric center changes position during the motion.

The FCM strategy is the most widespread and favored through the literature for two main reasons:

- The medical community widely recognizes that the best suturing results are achieved by following the natural curvature of the needle [8].
- Studies have shown that FCM results in less tissue damage compared to MCM [9].

Throughout the literature many researchers have

based their work on a FCM path algorithm. Hossein et al. [6] proposed a method where the needle follows a predefined semicircular trajectory, discretized into intermediate points. This approach reduces needle-tissue interaction and improves procedural precision. However, it is not fully autonomous, as it requires surgeon intervention to adjust the needle's orientation during insertion. A later improvement by the same authors introduced an optimization-based trajectory planning algorithm, ensuring the needle follows a circular arc with the same radius as its curvature, thereby adhering to the FCM principle [10]. Similarly, Sahba et al. [11] developed a path-planning algorithm based on the FCM strategy, incorporating a vision-based controller to estimate the needle's position. Instead, Cjow et al. developed an autonomous knot-tying method, useful for the stitch's closure [12].

1.1. Aim of the study

This study aims to further develop an optimization algorithm for estimating the needle's geometrical features and its center position to compute the FCM path in anastomosis procedures. A simulation environment is implemented to test the algorithm's performance and assess its robustness in various realistic scenarios. Finally, the effectiveness of the optimization process is evaluated.

2. Materials & Methods

The workflow of this study can be observed in Figure 2, where the key steps are presented.

Firstly, the adopted optimization model was developed by Pedram et al. [13] in 2017 for autonomous suturing via surgical robot through FCM algorithm. In their study, it is defined the 2D model of a wound to be sutured, scalable for different geometrical configurations, comprehensive of the main involved suture parameters. All the relevant aspects of the optimization model will be explained in Section 2.1.

After the problem definition, the algorithm is tuned by selecting the solution method and evaluating its precision (*grid sensitivity analysis*) to have reliable outputs. At this stage, considering a realistic anastomosis scenario from a geometrical point of view, the first optimization is executed to define the needle geometry and the path to be followed. Subsequently, the robustness of the algorithm is tested in the validation procedure (Section 3) for the second suture by integrating a *simulation environment* (Section 2.2) of the anastomosis setup with the aim of providing geometrical inputs for the optimization model. Finally, the suture parameters are compared (for different geometries) within two cases: optimizing the center position (OC) and fixing the center position based on the first outcome (FC).

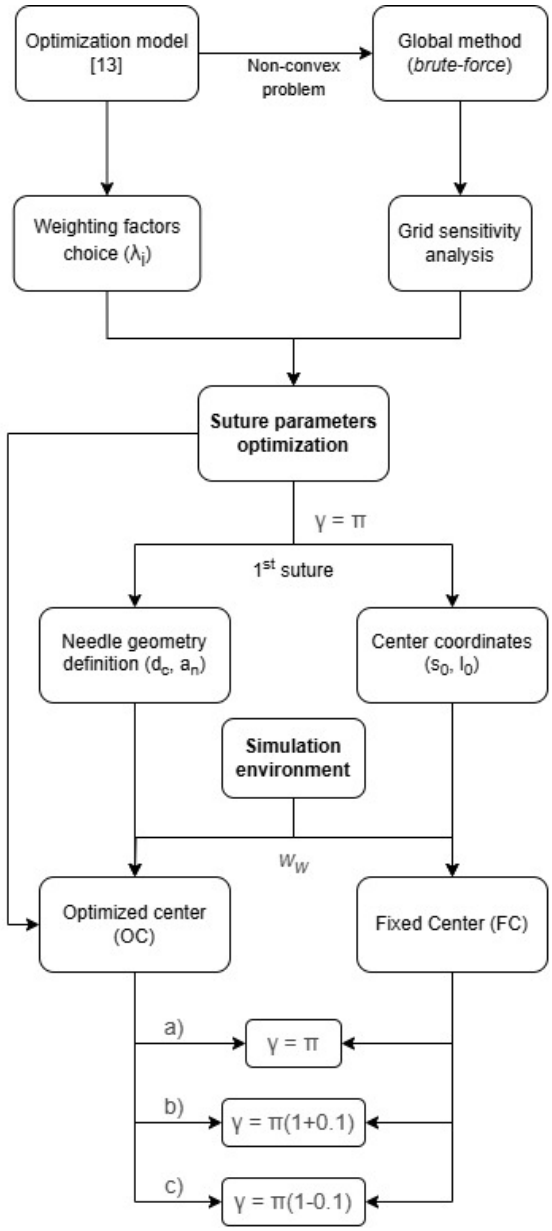


Figure 2: Project workflow

The technical aspects of the work are explained in three specific sections, respectively dedicated

to the optimization problem (2.1), to the development of the simulation environment in Pybullet, mimicking end-to-end anastomosis setup (2.2) and to their integration (2.3).

2.1. Suture Optimization

The anastomosis optimization model starts from the formulation of Pedram et al. [13] that defined the FCM path for a generic wound. In FCM, the trajectory of the needle is uniquely defined with the Equation (1). This expression describes the motion of the needle tip ($\vec{P}$) as it follows a circular trajectory around the fixed center ($\vec{C_0}$), with a radius determined by half the needle diameter ($\frac{d_c}{2}$). Finally, $\widehat{C_0P}$ is the unit vector that connects the needle center with the tip of the needle.

$$\vec{P(t)} = \vec{C_0} + \frac{d_c}{2}\widehat{C_0P(t)} \tag{1}$$

By looking at this formulation, the trajectory depends on the needle geometry (diameter d_c) and the center position ($\vec{C_0}$). Therefore, the optimization process has the aim to determine the best needle geometry and position for suturing, considering the specific wound configuration.

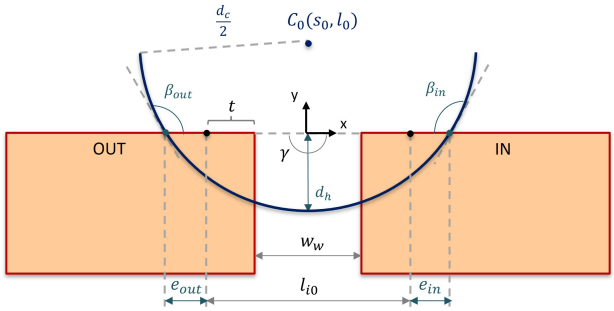


Figure 3: Geometric suture model

Figure 3 collects the main variables involved in the formulation of the problem that can be subdivided in outputs, inputs and suture parameters. The outputs are represented by the diameter (d_c), the coordinates of the center (s_0, l_0) and the needle shape (a_n) that refers to the fraction of circumference of the needle. The latter has only discrete values, based on commercially available shapes of $\frac{1}{4}$, $\frac{3}{8}$, $\frac{1}{2}$, $\frac{5}{8}$. The inputs are needed to define the tissue geometry and in this case are the tissue angle (γ), the wound width (w_w) and the desired entry-exit points distance (l_{i0}). Finally, the suture parameters are defined: entry and exit angle (β_{in} and β_{out}), entry and exit distances with respect to the desired points (e_{in} and e_{out}), the needle depth (d_h) and the needle-wound symmetry s_n (not visible in Figure 3) which is the absolute value of the x-coordinate of the center (s_0). The distance t is useful to define the suture parameters and it represents the line that links the vessel extremities to the ideal entry-exit points. Ideally, the angles β should be equal to $\frac{\pi}{2}$ for tissue-damage minimization [4], the distances l_{i0} and s_n should be zero and d_h is usually set to $\frac{l_{i0}}{2}$, but can be personalized: in this case its ideal value is defined equal to w_w because a large depth is not needed. These suture parameters form the basis to compute the optimal conditions (needle geometry and position). Their expression is obtained by pure geometrical considerations, knowing that the reference frame origin is placed in the midpoint of the two vessels (x-coordinate) in correspondence of the ideal entry-exit points (y-coordinate).

Cost function and kinematic constraints The cost function definition starts from the six suture parameters previously described (SP = β_{in}, β_{out}, e_{in}, e_{out}, d_h, s_n) and it is formulated to minimize the difference between the ideal values and the actual ones. The mathematical formulation is defined as follows:

$$\text{Minimize } f(s_0, l_0, d_c, a_n)$$

$$f = \sum_{i=1}^{6} \lambda_i \left|SP_{a,i} - SP_{d,i}\right| = \sum_{i=1}^{6} \lambda_i \Delta_i \tag{2}$$

where $SP_{a,i}$ and $SP_{d,i}$ are respectively the actual and desired suture parameters. The cost function minimizes the deviation between $SP_{a,i}$ and $SP_{d,i}$ with the addition of weighting factors (λ_i) that allow prioritization of certain parameters depending on surgical importance. The optimization problem is concluded by adding four constraints: *Bite time constraint*, *Extraction time constraint*, *Switching time constraints* and *needle depth constraint*. The first one is introduced to avoid tissue contact of the rear of the needle when the suture starts; analogously, the second ensures that the needle tip is not in contact with the tissue at the end of the suture. Then, the set of switching time constraints guarantees that there

exists an instant at which the needle is graspable at both ends.

The final constraint in Equation (3) is not present in the original work, but it is introduced to adapt the algorithm to an anastomosis case. Since the external diameter of the vessels is set to 10mm (as will be explained further in 2.2) the needle depth term d_h is restricted to a maximum value $d_{h,max} = 8mm$ in order to avoid the contact of the needle with the other side of the vessels.

$$d_h \leq d_{h,max} \tag{3}$$

Due to the fact that the expressions of the suture parameters are highly non-linear, the problem is probably not convex and the numerical methods for local minima can be unsuccessful for this application. For this purpose, a global method can be more suitable, in order to find the global minimum of the cost function. Due to the acceptable low amount of possible output values, the brute-force method is adopted, implemented in Python as *scipy.optimize.brute*. The output variables vary between predefined ranges:

- $(-\frac{l_{i0}}{2}; \frac{l_{i0}}{2})$ for the x-coordinate s_0.
- $(0; 2 \cdot l_{i0})$ for the y-coordinate l_0.
- (10;77) for the needle diameter (d_c).

Moreover, since the suture parameters are described in different units (mm for the distances and rad for the angles) the cost function (2) is not properly balanced for each variable. For this reason, a normalization is performed considering the maximum (Δ_{max}) and minimum (Δ_{min}) errors that can occur in the feasible set of solutions. Once a first optimization is run to obtain the Δ values, the effective optimization problem is updated with the following cost function:

$$\text{Minimize } \|f(s_0, l_0, d_c, a_n)\|$$

$$\|f\| = \frac{\sum_{i=1}^{6} \lambda_i \left(\frac{\Delta_i - \Delta_{i,\min}}{\Delta_{i,\max} - \Delta_{i,\min}} \right)}{\sum_{i=1}^{6} \lambda_i} = \frac{\sum_{i=1}^{6} \lambda_i \|\Delta_i\|}{\sum_{i=1}^{6} \lambda_i} \tag{4}$$

Normalization ensures that different suture parameters, which have different units (mm, rad), contribute proportionally to the total cost function, preventing any single term from dominating due to unit scaling.

Grid evaluation & λ-sensitivity analysis Finally, since the *brute-force* algorithm relies on grid discretization, a sensitivity analysis is conducted. In order to do that, some input parameters are kept fixed (l_{i0}, w_w, γ) and different number of grid elements are tested in order to have the best trade-off between accuracy and computational cost. The variation of the outputs between two consecutive grids is computed with the Equation (5).

$$\epsilon_i = \frac{\|grid_{i+1} - grid_i\|}{grid_i} \cdot 100 \tag{5}$$

The grids considered for the study are composed by 10, 20, 30, 40, 50 elements for each output variable. The Figure 4 shows the trend of the output variables as functions of the grid resolution (N). It is possible to qualitatively assess the solution convergence.

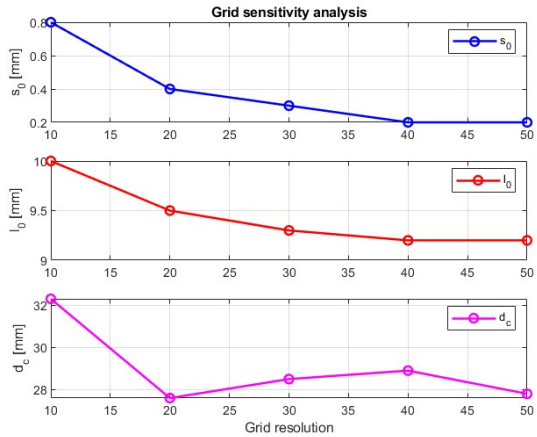


Figure 4: Solution convergence study

From a quantitative point of view, the Table 1 collects the corresponding variations computed with Equation (5), making reference to the values of the previous graph.

	Δs_0	Δl_0	Δd_c
ϵ_1	50	5.0	15
ϵ_2	25	2.1	3.3
ϵ_3	33	1.1	1.4
ϵ_4	0.0	0.0	3.8

Table 1: Grid sensitivity analysis

As can be seen in the last row, there are very small fluctuations in the results in correspondence of ϵ_4, which means that from the grid 4

(N=40) and the grid 5 (N=50) there are no substantial differences. For this reason, from now on the discretization grid refers to N=40. Once the grid is chosen, a further precision improvement is made to search a local minimum starting from the output of the global optimization, in order to have results not determined by the grid discretization. This process is implemented in the function *brute*, through the option '*finish=fmin*', where *fmin* stands for the *Nelder-Mead simplex algorithm*, able to find local minima without using the function gradient, which is very computationally demanding. The last step to properly set the optimization problem is the choice of the weighting factors for each suture parameter.

Before selecting all the values, a λ-sensitivity study is made to assess the impact of each suture parameter in the optimization output. As for the grid sensitivity study, the input parameters are kept fixed and six different simulations are performed, varying one weighting factor once. Alternatively, the λ_i factors are individually set to 0.5 (while keeping the others at 0.1), and the resulting optimization output values (X_{λ_i}) are compared with those from the default case (X_{λ_0}). In the default case $\lambda_0 = 1$, meaning all weighting factors are equal. A percentage difference is then calculated using Equation (6) to determine which input parameters have the greatest influence on the optimization results.

$$E_i = \frac{\|X_{\lambda_i} - X_{\lambda_0}\|}{X_{\lambda_0}} \cdot 100 \tag{6}$$

After the execution of the simulations, the following results are obtained:

- d_h and s_n influence the optimal results in a negligible manner.
- The diameter d_c is the most influenced output, followed by l_0.
- The distances e_{in} and e_{out} modify the y-coordinate of the center of about 17% and the diameter of more than 30%.
- The entry and exit angles β_{in} and β_{out} influence mostly the diameter (27%) than l_0 (3%).

Starting from these assessments, the weighting factors chosen for this work are resumed in Table 2, by considering that the aim of this work focuses on minimizing tissue damage, taking into account also the desired entry-exit points. Different sets of values can be chosen based on the surgeon's preference.

SP	β_{in}	β_{out}	e_{in}	e_{out}	d_h	s_n
λ_i	0.3	0.3	0.15	0.15	0.05	0.05

Table 2: Weighting factors λ

2.2. Simulation Environment

The surgical simulation is implemented using PyBullet (v.3.2.6), a physics engine that models interactions between rigid and deformable bodies in Python (v.3.9). The blood vessels are modeled as deformable bodies with an external radius of 10 mm and a vessel thickness of 1 mm, tyipical values of arteries [14]. The vessels are initially created in *.obj* format and later, through the use of the 'TetWild' repository [15], the mesh is changed from triangular to tetrahedral. Finally, the file is saved in *.vtk* format in accordance with PyBullet requirements for soft bodies. To simulate the typical deformation and elasticity of blood vessels, a Hookean model with Lamé and Cauchy constant values typical of arteries is used [16]. The needle is modeled as a rigid body imported from a *.stl* file, and its radius is defined through optimization. The needle movement is implemented such that it follows FCM, following Equation (1). Once the nth suture point is completed, the needle moves towards the position of the $n+1$ point, which is placed equidistant from the n point along the circumference of the vessel at an angle θ, defined in the Equation (7);

$$\theta = \frac{2\pi n_i}{n_t} \tag{7}$$

where n_i is the suture to be performed, and n_t is the total number of stitching points. In Figure 5 the simulation environment is shown, featured by the two vessels and the red points which symbolize the suture points.

During simulation, in real time, the user can change the speed of rotation of the needle during suturing and the position of the camera to have the best possible visualization of the surgical procedure through interactive sliders. The simulation is used to obtain the distances between the edges of the two vessels w_w, as shown in Figure 3, in the position of the suture $n+1$ during

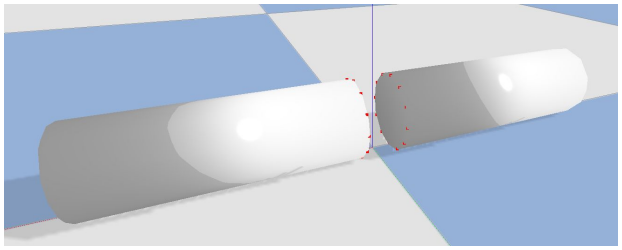


Figure 5: Simulation environment

the suture n (input parameter of the optimization problem). To obtain this information, visual markers are added to the nodes of the blood vessel mesh that are closer to the position of the suture $n+1$. The spatial coordinates of these markers are then extracted and the distance is computed as the Euclidean norm of the two position vectors (8).

$$\|\mathbf{d}\| = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2 + (z_1 - z_2)^2} \tag{8}$$

This information is then used for the validation process, as described in Section 3.

2.3. Integrating Optimization with Simulation Environment

Once the simulation environment is implemented, the last step is the linkage between the two different reference systems in order to execute the suture by correctly placing the needle center. Indeed, during optimization, the anastomosis scenario is treated as a 2D problem, where the wound geometry is projected onto the stitch plane. To transfer the 2D needle center coordinates (Local reference frame) into the 3D simulation environment (Global reference frame), a transformation matrix is needed. The Figure 6 shows a realistic scenario in which the optimization is performed for a generic stitch with a vertical inclination of θ, defined as in Equation (7).

Firstly, the local reference system is obtained through a rotation of $-\frac{\pi}{2}$ around the global z-axis, followed by an intrinsic rotation of $\frac{\pi}{2} - \theta$ around the new x-axis. Lastly, the translational component is defined in Equation (9) as a function of θ and the vessel radius (r).

$${}^{G}P = \begin{bmatrix} rsin(\theta) & 0 & r + rcos(\theta) \end{bmatrix}^{\mathsf{T}} \tag{9}$$

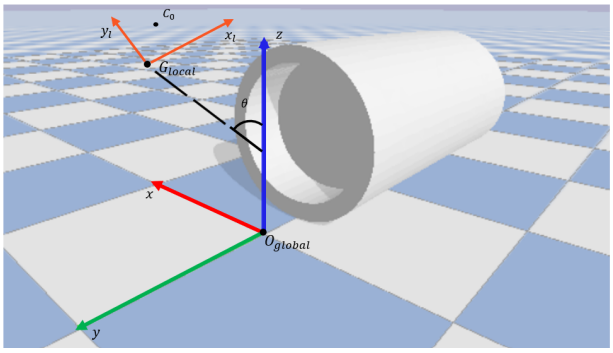


Figure 6: Global and local reference systems

At this stage, it is possible to write the transformation matrix ${}^{G}T_L$ that connects the two environments:

$${}^{G}T_L = \begin{bmatrix} 0 & sin(\theta) & -cos(\theta) & rsin(\theta) \\ -1 & 0 & 0 & 0 \\ 0 & cos(\theta) & sin(\theta) & r + rcos(\theta) \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{10}$$

This matrix converts the local coordinates frame of the optimization process into the global simulation frame. Finally, knowing that the needle center has coordinates defined by the optimization process in Equation (11), thanks to the transformation matrix in Equation (10), the coordinates of the needle center are easily obtained in the global reference frame through Equation (12).

$${}^{L}C_0 = \begin{bmatrix} s_0 & l_0 & 0 \end{bmatrix}^{\mathsf{T}} \tag{11}$$

$${}^{G}C_0 = {}^{G}T_L \cdot {}^{L}C_0 \tag{12}$$

Once the relationship between the algorithm and the simulation environment is established, the experimental phase begins.

3. Validation

To validate the optimization algorithm, a multi-step robustness study has been conducted. The first step is the same for all experiments and consists of optimizing both the needle parameters and its center. For this step, the initial wound width is set to $w_w = 5$ mm, the ideal entry-exit distance is fixed at $3w_w = 15$mm and the angle between the vessels is $\gamma = \pi$. The needle geometry obtained through the optimization (d_c, a_n)

is kept fixed for the next studies. Then, for the following sutures, two scenarios are analyzed:

1. The needle center is computed for each suture stitch, adopting the needle geometry previously determined (optimized center).
2. The needle follows the trajectory by using all the output parameters obtained in the first suture process, scaled to the next stitch positions, avoiding any further optimization (fixed center).

Moreover, to simulate realistic conditions, also the overall geometrical configuration is changed through the tissue angle (γ), input parameter that can change during anastomosis. In this study, three different possibilities are considered:

a) The angle between the vessels remains constant: $\gamma = \pi$.
b) The angle between the vessels increases by 10%: $\gamma = \pi(1 + 0.1)$.
c) The angle between the vessels decreases by 10%: $\gamma = \pi(1 - 0.1)$.

Additionally, in all the considered cases, the distance between the vessels is computed through the simulation environment. The Euclidean norm, shows in Equation (8), is used to compute the new input parameter w_w. For each suture, different values of w_w have been used to obtain results that are less affected by simulation variability. The computed suture parameters are then averaged, and their standard deviation is calculated to account for variability. The final values are reported in the tables presented in Section 4. After completing the experiments, the suture parameters are compared in terms of absolute difference between the real values obtained from the simulations and the ideal ones, as shown in Equation 13. This metric is used for the following suture parameters (SP) : β_{in}, β_{out}, e_{in}, e_{out} and d_h. The error is defined as:

$$e = |SP_{id} - SP_{sim}| \tag{13}$$

where SP_{id} is the ideal value of one of the suture parameters, reported in the Section 2.1, and SP_{sim} is the value computed with the informations taken from both the simulation and the optimization. These experiments help to determine whether optimizing the needle center for each suture point —although more time consuming— results in a more accurate suturing process, or if optimizing only the first suture point is a sufficiently good option, despite the potential lack of accuracy.

4. Results

As previously described, for the first suture the input parameters (γ, w_w, l_{i0}) have been chosen and the optimization problem is solved by considering all the possible needle shapes. In Figure 7 it is reported the typical output of the optimization process for each a_n value: in this case it is presented only the optimal needle shape ($a_n = 0.5$). In particular the Figure 7a collects the cost function value for each possible combination of center coordinates (s_0, l_0) and diameter (d_c) within the feasible set, taking into account the constraints. The red marker indicates the optimal set of outputs which is surrounded by discretized points at which the cost function assumes its minimum values. As expected, by increasing the needle diameter, the y-coordinate of the center must be higher and the cost function value increases as well, worsening the suture parameters errors. For what concerns the Figure 7b, it is shown a visible representation of the needle geometry and its center.

From this outcome the needle is selected and the starting point for the experiments is defined. In the table 3, the optimal values for the first suture are resumed, useful for the subsequent steps.

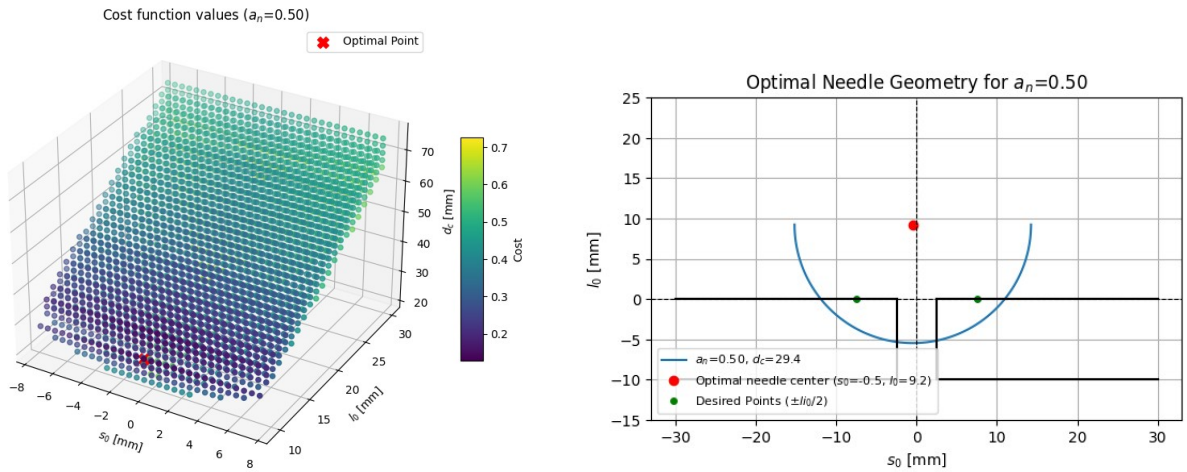
Output	s_0	l_0	d_c	a_n
	± 0.5mm	9.2mm	29mm	0.5

Table 3: Optimal outcome for the first suture

Starting from this, the suture parameters and the needle's center have been computed for each suture with different values of w_w, obtained after some trials in the simulation environment (4.2 mm, 5.4 mm, 6.9 mm). In the following tables only the results for the second suture are described, but the same can easily be extended to all the sutures.

4.1. Case a: $\gamma = \pi$

In Table 4, the suture parameters (SP) for both the optimized center (OC) and for the fixed center (FC) are presented for the *case a*. The re-



(a) Cost function value distribution

(b) Optimal needle geometry and position

Figure 7: Optimal results for the first suture ($a_n = 0.5$)

ported suture parameters correspond to the average value of the suture parameters obtained with the w_w simulated values with the associated standard deviation.

SP	Ideal	OC_a	FC_a
$\beta_{in}[rad]$	1.57	2.26 ± 0.01	2.25 ± 0.01
$\beta_{out}[rad]$	1.57	2.26 ± 0.01	2.25 ± 0.01
$e_{in}[mm]$	0	2.94 ± 2.03	2.45 ± 2.03
$e_{out}[mm]$	0	2.94 ± 2.03	3.45 ± 2.03
$d_h[mm]$	5	5.30 ± 0.01	5.30 ± 0.0

Table 4: Suture parameters for $\gamma = \pi$

In *case a* the computed optimized center (OC_a) is $s_0 = 0.0$ mm and $l_0 = 9.2 \pm 0.01$ mm. Analyzing the results in Table 4, it is evident that when γ remains constant, the optimization process has a minimal effect on the suture parameters. Both β_{in} and β_{out} remain nearly unchanged between OC and FC, as well as for the entry and exit errors, e_{in} and e_{out}, which show small variations. This implies that for scenarios where the tissue angle remains stable, performing optimization at each step may not be necessary in a real surgical setting, as the improvement does not justify the additional computational time.

4.2. Case b: $\gamma = \pi(1 + 0.1)$

In Table 5, the suture parameters for *case b* are presented for both the OC and the FC. In this experiment the needle center is placed at $s_0 = 0.0mm$ and $l_0 = 9.7 \pm 0.3mm$.

SP	Ideal	OC_b	FC_b
$\beta_{in}[rad]$	1.57	2.42 ± 0.01	2.36 ± 0.03
$\beta_{out}[rad]$	1.57	2.42 ± 0.01	2.36 ± 0.03
$e_{in}[mm]$	0	2.97 ± 2.05	2.83 ± 2.33
$e_{out}[mm]$	0	2.97 ± 2.05	3.98 ± 2.32
$d_h[mm]$	5	3.90 ± 0.10	4.42 ± 0.21

Table 5: Suture parameters for $\gamma = \pi(1 + 0.1)$

For *case b*, the optimized center (OC_b) results in better exit positioning, as evidenced by the lower e_{out} value compared to the FC case. However, the entry and exit angles are slightly worse than in the FC case. This suggests that while optimization may improve specific parameters, it does not always lead to a uniform enhancement across all suture parameters. In a real surgical environment, a surgeon might prioritize different aspects of the suture depending on the specific needs of the anastomosis that is being performed, and the optimization algorithm should be adjusted accordingly to balance precision and efficiency, thus meeting the surgeon's needs.

4.3. Case c: $\gamma = \pi(1 - 0.1)$

The results of the last experiment, *case c*, are displayed in Table 6. The center coordinates (OC_c) are $s_0 = 0.0mm$ and $l_0 = 8.7 \pm 0.30mm$.

SP	Ideal	OC_c	FC_c
$\beta_{in}[rad]$	1.57	2.10 ± 0.01	2.13 ± 0.02
$\beta_{out}[rad]$	1.57	2.10 ± 0.01	2.13 ± 0.02
$e_{in}[mm]$	0	2.97 ± 2.05	2.19 ± 1.80
$e_{out}[mm]$	0	2.97 ± 2.05	3.07 ± 1.80
$d_h[mm]$	5	6.60 ± 0.01	6.04 ± 0.40

Table 6: Suture parameters for $\gamma = \pi(1 - 0.1)$

For *case c*, the optimized center leads to noticeable improvements in the entry and exit angles. This is particularly relevant in real-world suturing, as better angles reduce tissue stress and damage, improving suture durability. Therefore, when γ is decreasing, optimization at each step may become more relevant to minimize tissue trauma and enhance surgical precision, therefore justifying a longer procedure due to a higher computational cost.

5. Conclusions

Nowadays, the transition from robotic-assisted surgery to fully autonomous robotic procedures is becoming increasingly relevant. This work aims to automate a repetitive task such as suturing. To achieve this, a path-planning algorithm is developed in which the needle follows a circular trajectory around a fixed center. An optimization problem is formulated to determine the optimal position of this fixed center, minimizing tissue trauma and improving surgical outcomes. Several realistic simulations are conducted to evaluate the efficiency of center optimization for each stitch. Based on the simulation results, reported in Section 4, it can be concluded that the necessity of optimization for each suture strongly depends on the geometry of the surgical scenario.

Case *a*: If the vessel angle remains at π, the center coordinates remain stable, making optimization at each suture unnecessary, even if vessel distance varies. For instance, in the condition where the tissue angle does not vary, it can be seen from Table 4 that both β_{in} and β_{out} are very similar in the OC and FC case. Instead there is a slight change in e_{in} and e_{out}; the former is worsened by 0.49 mm in the OC situation while the latter is improved of 0.51 mm. These very small variations of the suture parameters do not justify the loss of time needed for the optimization, which is in the order of 10 seconds.

Case *b*: If the optimization is performed there is an improvement of 1.01 mm for the exit point of the suture, while the entry point remain rather untouched. The entry and exit angles, however, are slightly worsened ($0.08rad$) with respect to the FC case. This behavior can be explained because the cost function encompasses all the suture parameters and, if the surgeon would like to have a focus on the angles rather than the entry/exit points, different weighting factors should be given to the cost function in order to better achieve the surgeon's request.

Case *c*: The y-coordinate of the needle center varies significantly. Optimizing at each suture improves entry and exit angles, reducing tissue damage, though at the cost of slight inaccuracies in entry and exit points. To conclude, from these results it can be assessed that if the tissue angle γ remains equal or it increases between sutures is not convenient to optimize the needle center at each step, because acceptable results can be achieved with the FC case. If between sutures γ tends to reduce, optimization at each suture can be useful since better entry and exit angles are obtained, leading to a less severe tissue damage.

These findings provide valuable insights that could enhance real-world surgical procedures, particularly in microsurgery where precision is crucial. Understanding how the tissue angle γ influences optimization decisions could help surgeons and robotic systems adapt their suturing strategies dynamically. In practice, this means that automated or robotic-assisted suture placement could be programmed to selectively optimize needle positioning based on the local tissue geometry, improving efficiency. However, scaling this approach to real-world systems presents several challenges. Real biological tissues, for example, exhibit greater variability in elasticity and deformation compared to simulations, requiring more sophisticated models to ensure accurate predictions and angle assessement maybe difficult to be performed. In addition to this, real-time adaptation of optimization parameters

in a surgical setting would requires fast computational methods and a flawless integration with robotic systems or surgeon-assistive technologies. Finally, validation in experimental and clinical settings would be necessary to confirm that the predicted benefits, such as reduced tissue damage and improved suture quality, translate into measurable improvements in patient outcomes.

5.1. Limitations & Future Developments

This study presents some limitations in the methodologies and the tools used for the simulations. Firstly, the functionality of Pybullet library for soft bodies is very limited and not optimal for this specific anastomosis application. In particular, the environment allowed to simulate the physical interaction between the needle and the vessels, however is not possible to anchor the bodies together at a specific suture point, to simulate the stitching procedure. For this reason, the computed distance in the validation procedure (w_w) oscillates between the initial value and does not vary as in real suture procedure would do. Even if this distance has a reduced impact on the optimization results, this simplification could be avoided by implementing the anastomosis scenario in more complex environments as, for instance, the software SOFA (Simulation Open Framework Architecture). Using a simulation software like SOFA would provide more advanced soft body dynamics and better handling of biomechanical interactions, making it a promising alternative to PyBullet. Its key advantages are:

- Ability to define and control soft body deformations more precisely when modelling biological tissues.
- Support of anchoring mechanisms that allow sutures to be realistically simulated, meaning that vessels can be stitched together and held in place throughout the simulation, providing an accurate representation of the physical constraints.
- Real-time interaction with deformable structures, making it more suitable for integrating robotic manipulators in future studies.

Indeed in this study no manipulators have been implemented neither in the simulation nor in a practical experiment. For this reason, in subsequent studies it could be reasonable to introduce robotic arms in order to test the different scenarios and verify the simulation results in real experimental cases, checking also the feasibility of the suture operations in the microsurgery environment.

Another aspect not explored in this study is the role of the angular velocity of the needle. In the simulations, the user is given the opportunity to modify this value, which is then kept constant throughout the procedure. A potential future improvement could involve studying how the rotational speed affects tissue damage, aiming to determine the optimal value that minimizes trauma. It could be considered in the optimization algorithm, by making parameters dependent on time.

Finally, regarding the optimization algorithm itself, it is possible to tune the model based on the most relevant aspects for a specific surgery procedure. As a first variation, the weighting factors can be modulated depending on the specific clinic case and based on the surgeon's preferences. Moreover, further modifications can be applied to the geometrical model, either by considering additional suture parameters or completely neglecting some of them to better meet the patient-specific clinical requirements.

References

[1] Ondine Cleaver and Paul A. Krieg. Vascular development. In Nadia Rosenthal and Richard P. Harvey, editors, *Heart Development and Regeneration*, volume 1, pages 85–102. Academic Press, 2010.

[2] François Saade, Alexandre Quemener-Tanguy, Laurent Obert, Sebastien El-Rifai, Camille Bouteille, and François Loisel. Tricks in end-to-end anastomosis in microsurgery: a systematic review. *Indian Journal of Surgery*, 84(4):740–747, 2022.

[3] Taseen F. Haque, J. Everett Knudsen, Jonathan You, Alvin Hui, Hooman Djaladat, Runzhuo Ma, Steven Cen, Mitchell Goldenberg, and Andrew J. Hung. Competency in robotic surgery: Standard setting for robotic suturing using objective assessment and expert evaluation. *Journal of Surgical Education*, 81(3):422–430, 2023.

[4] Taseen F. Haque, Alvin Hui, Jonathan You, Runzhuo Ma, Jessica H. Nguyen, Xiaomeng Lei, Steven Cen, Monish Aron, Justin W. Collins, Hooman Djaladat, et al. An assessment tool to provide targeted feedback to robotic surgical trainees: Development and validation of the end-to-end assessment of suturing expertise (ease). *Urology Practice*, 9(6):532–539, 2022.

[5] Leo Meller Zih-Yun Chiu Michael Yip Ryan K. Orosco4 Benjamin T. Ostrander, Daniel Massillon. The current state of autonomous suturing: a systematic review. *Surgical endoscopy*, 2024.

[6] Dmitry Oleynikov Benjamin Terry Hossein Dehghani, Shane Farritor. Automation of suturing path generation for da vinci-like surgical robotic systems. In *Proceedings of the Design of Medical Devices Conference (DMD)*, page V001T07A008. ASME, 2018.

[7] Yonjae Kim Axel Krieger Simon Leonard, Kyle L. Wu and Peter C.W. Kim. Smart tissue anastomosis robot (star): A vision-guided robotics system for laparoscopic suturing. *IEEE TRANSACTIONS ON BIOMEDICAL ENGINEERING*, 2014.

[8] B. Pulles F. Kappelhof J. Ruurda, I. Broeders and C. Van der Werken. "manual robot assisted endoscopic suturing: time-action analysis in an experimental model". *Surgical endoscopy and other interventional techniques*, 2004.

[9] C. Doignon F. Nageotte, P. Zanne and M. De Mathelin. "stitching planning in laparoscopic surgery: Towards robot-assisted suturing". *Int'l Journal of Robotics Research*, 2008.

[10] Lou Cubrich Dmitry Oleynikov Shane Farritor Hossein Dehghani, Yue Sun and Benjamin Terry. An optimization-based algorithm for trajectory planning of an under-actuated robotic arm to perform autonomous suturing. *IEEE Transactions on Robotics*, 2020.

[11] Pezhman Mardanbeigi Sahab Pedram and Pierre Dupont. Autonomous suturing framework and quantification using a cable-driven surgical robot. *IEEE Transactions on Robotics*, 40(1):170–202, 2024.

[12] D.-L. Chow, R. C. Jackson, M. C. Cavusoglu, and W. Newman. A novel vision guided knot-tying method for autonomous robotic surgery. In *2014 IEEE International Conference on Automation Science and Engineering (CASE)*, pages 504–508, 2014.

[13] Sahba Aghajani Pedram, Peter Ferguson, Ji Ma, Erik Dutson, and Jacob Rosen. Autonomous suturing via surgical robot: An algorithm for optimal selection of needle diameter, shape, and path. In *2017 IEEE International conference on robotics and automation (ICRA)*, pages 2391–2398. IEEE, 2017.

[14] Franco Montevecchi Alberto Redaelli. *Biomeccanica: analisi multiscala di tessuti biologici*. Pàtron, 2019.

[15] Yixin Hu. Tetwild. `https://github.com/Yixin-Hu/TetWild`, 2018. GitHub repository.

[16] COMSOL Multiphysics. Fluid structure interaction in a network of blood vessels. *Structural mechanics module model library manual*, 1, 2014.